Homework 7

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Problem 1

Let $\mathcal{V}$ be a linear space over $\mathbb{F}$, $\mathcal{A} \in L(\mathcal{V})$, $\mathcal{W} \subset \mathcal{V}$ be $\mathcal{A}$-invariant, and $\psi(\lambda)$ be the minimal polynomial of $\mathcal{A}$ at $\mathcal{W}$. Show that there exists $x \in \mathcal{W}$ such that the minimal polynomial of $\mathcal{A}$ at x is $\psi(\lambda)$.

Solution

If $\mathcal{W} = \{0\}$, take $x = 0$ and $\psi = 1$. Otherwise write $\psi = \prod_{i=1}^{r} p_i^{m_i}$ with distinct monic irreducibles p_i, and put $q_i = \psi / p_i^{m_i}$. Minimality of ψ provides $v_i \in \mathcal{W}$ with $(\psi/p_i)(\mathcal{A})v_i \neq 0$. Set $y_i = q_i(\mathcal{A})v_i \in \mathcal{W}$. Then

$$p_i(\mathcal{A})^{m_i} y_i = 0, \qquad p_i(\mathcal{A})^{m_i - 1} y_i \neq 0.$$

Thus the minimal polynomial at y_i is $p_i^{m_i}$. These polynomials are pairwise coprime, so the coprime-sum property gives

$$\boxed{x = \sum_{i=1}^{r} y_i \in \mathcal{W}, \qquad m_x = \prod_{i=1}^{r} p_i^{m_i} = \psi}.$$

Problem 2

Let $\mathcal{A} \in L(\mathbb{C}_n[\lambda])$ be defined as

$$\mathcal{A}[f(\lambda)] = f(\lambda + a), \qquad \forall f(\lambda) \in \mathbb{C}_n[\lambda].$$

for a nonzero constant $a \in \mathbb{C}$; its matrix representation A with respect to the basis $1, \lambda, \lambda^2, \ldots, \lambda^n$ was found in Homework 5. Find out

(1) the Smith canonical form S of the eigenmatrix $\lambda E - A$, and the Jordan canonical form J of A;

(2) the transition matrix X that reduces A to the Jordan canonical form J, i.e. $X^{-1}AX = J$.

Solution

(1) Write the polynomial variable as s and put $\Delta = \mathcal{A} - I$. It lowers nonconstant polynomial degrees, with $\Delta^{n+1} = 0$ and $\Delta^n(s^n) = n!a^n \neq 0$. Hence there is one Jordan block of size $n + 1$:

$$\boxed{S = \operatorname{diag}(I_n, (\lambda - 1)^{n+1}), \qquad J = J_{n+1}(1)}.$$

(2) Take the following polynomials and use their coordinates in $(1, s, \ldots, s^n)$ as the columns of X, indexed from zero:

$$f_0(s) = 1, \qquad f_k(s) = \frac{1}{k!a^k} \prod_{r=0}^{k-1} (s - ra) \quad (1 \leq k \leq n).$$

$$\boxed{X_{ik} = [s^i] f_k(s) \quad (0 \leq i, k \leq n)}, \qquad X_{kk} = \frac{1}{k!a^k} \neq 0.$$

Here $[s^i]$ extracts a coefficient. The upper triangular X is invertible. Since $f_k(s + a) = f_k(s) + f_{k-1}(s)$, with $f_{-1} = 0$, we have $AX = XJ$ and hence $X^{-1}AX = J$.

Problem 3

Let E be an identity matrix with appropriate dimension,

$$\lambda+3, \quad \lambda+3, \quad (\lambda+3)^2, \quad \lambda+2+5j, \quad \lambda+2-5j,$$
$$(\lambda+2+5j)^2, \quad (\lambda+2-5j)^2.$$

are elementary divisors for $\lambda E - A$.

(1) Please find out the Smith canonical form S for $\lambda E - A$, the natural normal form N and the Jordan canonical form J for A.

(2) Let $\lambda_1 = -3$, $\lambda_2 = -2+5j$, $\lambda_3 = -2-5j$ and

$$T(\lambda_i) \triangleq \{x : \exists k \in \mathbb{Z}_+,\ (A-\lambda_i E)^k x = 0\}.$$

$\mathbb{Z}_{+,0}$ is the set of non-negative integers. Show that $T(\lambda_i) = \ker[(A-\lambda_i E)^2]$, $i = 1,2,3$, and determine the corresponding dimensions $\dim T(\lambda_i)$, $i = 1,2,3$.

Solution

Use $j^2 = -1$. Let $J_m(\rho)$ have diagonal ρ and superdiagonal ones; $C(p)$ is the column companion matrix, with subdiagonal ones and negated coefficients of monic p in ascending order in its last column.

(1) Put $x = \lambda+3$, $q = (\lambda+2)^2+25$. Sorting exponents and padding on the left gives nonconstant invariants x, xq, x^2q^2, of degrees one, three and six. Thus

$$\boxed{S = \operatorname{diag}(I_7, x, xq, x^2q^2), \qquad N = \operatorname{diag}(-3, C(xq), C(x^2q^2))}.$$

Write $K_\pm(a,b) = \operatorname{diag}(a \pm bj, J_2(a \pm bj))$. Then

$$\boxed{J = \operatorname{diag}(-3, -3, J_2(-3), K_-(-2,5), K_+(-2,5))}.$$

(2) At each eigenvalue the largest Jordan block has size two, so $(A-\lambda_i I)^2$ vanishes on its generalized eigenspace. The operator $A-\lambda_i I$ is invertible on all other components. Therefore

$$\boxed{T(\lambda_i) = \ker(A-\lambda_i I)^2, \qquad (\dim T(\lambda_1), \dim T(\lambda_2), \dim T(\lambda_3)) = (4,3,3)}.$$

Problem 4

Given a matrix A whose elementary divisor are

$$(\lambda+1),\quad (\lambda+1)^2,\quad (\lambda+1+j),\quad (\lambda+1-j),$$
$$(\lambda+1)^2,\quad (\lambda+1+j)^2,\quad (\lambda+1-j)^2.$$

(1) Find out the Smith canonical form S for $\lambda E - A$;

(2) Find out the Natural normal form N, the complex Jordan canonical form J_{complex} and the real Jordan canonical form J_{real} for A;

(3) The eigenvalues of matrix A are denoted as $\lambda_1 = -1$, $\lambda_2 = -1-j$, $\lambda_3 = -1+j$. For λ_i, $i = 1, 2, 3$, give the geometric multiplicity q_i and algebraic multiplicity $\widehat{n}_i$;

(4) Find out $\dim T(\lambda_i)$, $i = 1, 2, 3$, where

$$T(\lambda_i) \triangleq \{x : \exists k \in \mathbb{Z}_+,\ (A - \lambda_i E_n)^k x = 0\}.$$

Solution

Put $x = \lambda + 1$, $q = x^2 + 1$ and retain the block notation from Problem 3.

(1) The nonconstant invariants are x, x^2q, x^2q^2, of total degree eleven, so

$$\boxed{S = \operatorname{diag}(I_8, x, x^2q, x^2q^2)}.$$

(2)

$$\boxed{N = \operatorname{diag}(-1, C(x^2q), C(x^2q^2))},$$

$$\boxed{J_{\text{complex}} = \operatorname{diag}(-1, J_2(-1), J_2(-1), K_-(-1,1), K_+(-1,1))}.$$

Put $R = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$ and $R_2 = \begin{pmatrix} R & I_2 \\ 0 & R \end{pmatrix}$. Pairing conjugate blocks gives

$$\boxed{J_{\text{real}} = \operatorname{diag}(-1, J_2(-1), J_2(-1), R, R_2)}.$$

(3) Geometric multiplicity counts blocks, while algebraic multiplicity sums their sizes:

λ_i	q_i	$\widehat{n}_i$
-1	3	5
$-1-j$	2	3
$-1+j$	2	3

(4) Generalized eigenspace dimensions equal algebraic multiplicities, hence

$$\boxed{\dim T(\lambda_1) = 5, \qquad \dim T(\lambda_2) = 3, \qquad \dim T(\lambda_3) = 3}.$$